Second Order ODEs

The **general form** of a second order ODE is

$$\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} = f(t, y, \frac{\mathrm{d}y}{\mathrm{d}t})$$

for some given function f.

1 Linear Homogeneous Equations

Definition 1.1 (Linear Equation)**.**

A second order ODE is called **linear** if it can be written in the form

$$a(t) \cdot y'' + b(t) \cdot y' + c(t) \cdot y = g(t)$$

If $a(t) \neq 0$, we can divide both sides by $a(t)$ to get $y'' + p(t) \cdot y' + q(t) \cdot y = r(t)$.

Definition 1.2 (Homogeneous Equation)**.**

A linear equation is called **homogeneous** if it can be written in the form

$$a(t) \cdot y'' + b(t) \cdot y' + c(t) \cdot y = 0$$

Otherwise, it is called **non-homogeneous**.

Theorem 1.1 (Existence and Uniqueness Theorem)**.**

Consider the IVP:

$$y'' + p(t) \cdot y' + q(t) \cdot y = r(t), \qquad y(t_0) = y_0, \quad y'(t_0) = y_1$$

if there exists an open interval $I \subseteq \mathbb{R}$ such that $t_0 \in I$ and p, q and r are **continuous** on I, then there is a **exact one** solution $y = \phi(t)$, and it exists **throughout** I.

Theorem 1.2 (Principle of Superposition)**.**

If y_1 and y_2 are two solutions of the **homogeneous equation**

$$a(t) \cdot y'' + b(t) \cdot y' + c(t) \cdot y = 0$$

then any linear combination $c_1 y_1(t) + c_2 y_2(t)$ is also a solution of the ODE.

Definition 1.3 (Wronskian)**.**

The **Wronskian**, or **Wronskian determinant**, of two functions y_1 and y_2 is defined as

$$W(y_1, y_2)(t) := \begin{vmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{vmatrix} = y_1(t) \cdot y_2'(t) - y_2(t) \cdot y_1'(t)$$

Theorem 1.3 (Abel's Theorem).

Suppose $p(t)$ and $q(t)$ are continuous on an open interval I, and y_1 and y_2 are two non-zero solutions of the ODE

$$y'' + p(t) \cdot y' + q(t) \cdot y = 0 \tag{1}$$

Then, the Wronskian of y_1 and y_2 is given by

$$W(y_1, y_2)(t) = c \cdot \exp\left(-\int p(t)\,\mathrm{d}t\right) \qquad \text{where } c \in \mathbb{R} \text{ is a constant depending on } y_1, y_2$$

Remark. If $y(t), p(t), q(t) \in \mathbb{C}$, then $c \in \mathbb{C}$.

Corollary $\quad \exists t_0 \in I \text{ s.t. } W(y_1, y_2)(t_0) \neq 0 \implies \forall t \in I,\ W(y_1, y_2)(t) \neq 0$

Proof (Abel's Theorem).

$$\begin{aligned}
\begin{cases} y_2 y_1'' + y_2 p y_1' + y_2 q y_1 = 0 \\ y_1 y_2'' + y_1 p y_2' + y_1 q y_2 = 0 \end{cases} &\implies (y_2 y_1'' - y_1 y_2'') + p \cdot (y_2 y_1' - y_1 y_2') = 0 \\
&\implies W' + p \cdot W = 0 \\
&\implies W = c \cdot \exp\left(-\int p(t)\,\mathrm{d}t\right) \quad \text{for some constant} \quad c
\end{aligned}$$

□

Theorem 1.4 (Linear Dependence and Wronskian).

Suppose y_1 and y_2 are two solutions of an ODE in the form of eq. (1). Then,

$$y_1(t) \quad \text{and} \quad y_2(t) \quad \text{are } \textbf{linearly independent} \text{ on} \quad t \in I \qquad \iff \qquad \forall t \in I,\ W(y_1, y_2)(t) \neq 0$$

***Proof*.**

$$\begin{aligned}
\text{linear independence} \quad &\iff \quad \forall c_1, c_2 \in \mathbb{C},\ \Big(c_1 y_1(t) + c_2 y_2(t) = 0\Big) \to \Big(c_1 = c_2 = 0\Big) && \text{[Proposition 1]} \\
&\iff \quad \forall c_1, c_2 \in \mathbb{C},\ \begin{bmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \to \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} && \text{[Proposition 2]} \\
&\iff \quad W(y_1, y_2)(t) \not\equiv 0 \\
&\iff \quad \forall t \in I,\ W(y_1, y_2)(t) \neq 0 \quad \text{by the corollary of Abel's Theorem}
\end{aligned}$$

([Proposition 2] $\implies$ [Proposition 1]) is obtained by combining [Proposition 2] with the following fact:

$$\forall c_1, c_2 \in \mathbb{C},\ \Big(c_1 y_1(t) + c_2 y_2(t) = 0\Big) \to \Big(c_1 y_1'(t) + c_2 y_2'(t) = 0\Big)$$

□

Theorem 1.5.

Suppose $p(t)$ and $q(t)$ are continuous on an open interval I, and y_1 and y_2 are two solutions of the ODE

$$y'' + p(t) \cdot y' + q(t) \cdot y = 0 \quad \text{for} \quad t \in I \tag{2}$$

Let $\phi(t)$ be the **unique** solution (by Theorem 1.1) of the IVP:

$$y'' + p(t) \cdot y' + q(t) \cdot y = 0 \qquad y(t_0) = m, \quad y'(t_0) = n \quad \text{for} \quad t \in I \quad \text{and} \quad t_0 \in I \tag{3}$$

Then, for $t \in I$,

$$\exists!\, c_1, c_2 \in \mathbb{R} \text{ s.t. } \phi(t) = c_1 y_1(t) + c_2 y_2(t) \qquad \iff \qquad W(y_1, y_2)(t_0) \neq 0 \tag{4}$$

Remark. In the assumption, y_1 and y_2 are not required to be linearly independent. Thus c_1, c_2 are required to be unique in the left hand side of eq. (4).

Proof.

Sufficiency. Since $\phi(t_0) = m$ and $\phi'(t_0) = n$, we have

$$\begin{aligned} \exists!\, c_1, c_2 \in \mathbb{R} \text{ s.t. } \phi(t) = c_1 y_1(t) + c_2 y_2(t) \quad &\implies \quad \exists!\, c_1, c_2 \in \mathbb{R} \;\text{ s.t. } \begin{bmatrix} y_1(t_0) & y_2(t_0) \\ y_1'(t_0) & y_2'(t_0) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} m \\ n \end{bmatrix} \\ &\implies \quad \exists b_1, b_2 \in \mathbb{R},\ \exists!\, c_1, c_2 \in \mathbb{R} \;\text{ s.t. } \begin{bmatrix} y_1(t_0) & y_2(t_0) \\ y_1'(t_0) & y_2'(t_0) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \\ &\iff \quad W(y_1, y_2)(t_0) \neq 0 \end{aligned}$$

Necessity.

$$W(y_1, y_2)(t_0) \neq 0 \quad \implies \quad \exists!\, c_1, c_2 \in \mathbb{R} \;\text{ s.t. } \begin{bmatrix} y_1(t_0) & y_2(t_0) \\ y_1'(t_0) & y_2'(t_0) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} m \\ n \end{bmatrix}$$

Let $\varphi(t) := c_1 y_1(t) + c_2 y_2(t), t \in I$. Then we have $\varphi(t_0) = m$ and $\varphi'(t_0) = n$.

$$\begin{aligned} \text{Theorem 1.2 (Principle of Superposition)} \quad &\implies \quad \varphi(t) \text{ is a solution of the ODE eq. (2).} \\ \text{Theorem 1.1 (Existence and Uniqueness Theorem)} \quad &\implies \quad \varphi(t) \text{ is the } \textbf{only} \text{ solution of the IVP in eq. (3).} \end{aligned}$$

Thus, $\varphi(t) = \phi(t)$ and we have

$$W(y_1, y_2)(t_0) \neq 0 \quad \implies \quad \exists!\, c_1, c_2 \in \mathbb{R} \text{ s.t. } \phi(t) = \varphi(t) = c_1 y_1(t) + c_2 y_2(t)$$

□

Theorem 1.6.

Suppose $p(t)$ and $q(t)$ are continuous on an open interval I, and y_1 and y_2 are two solutions of ODE

$$y'' + p(t) \cdot y' + q(t) \cdot y = 0$$

for $t \in I$. Then,

$$\text{The } \textbf{general solution} \text{ is given by } \phi(t) = c_1 y_1(t) + c_2 y_2(t). \quad \iff \quad \exists t_0 \in I \text{ s.t. } W(y_1, y_2)(t_0) \neq 0.$$

y_1, y_2 are said to form a **fundamental set of solutions (FSS)** of the ODE, if their Wronskian is non-zero.

Remark. Theorem 1.6 is used to determine whether a given pair of solutions y_1, y_2 form a FSS, rather than verifying the existence and form of solutions. An ODE satisfying the condition (continuous coefficients) always has solutions, and the solutions form a 2D vector space. By Theorem 1.4 (Linear Dependence and Wronskian), the linear independence of y_1, y_2 also indicates that they form a FSS.

Proof.

Let S denote the set of all solutions of the ODE. We aim to prove that

$$\forall y \in S, \exists c_1, c_2 \in \mathbb{R} \text{ s.t. } y(t) = c_1 y_1(t) + c_2 y_2(t) \quad \iff \quad \exists t_0 \in I \text{ s.t. } W(y_1, y_2)(t_0) \neq 0$$

Let $\mathcal{S}_{\texttt{IVP}}[t^*, m, n]$ denote the solution of the following IVP (by Theorem 1.1, there is exactly one solution):

$$y'' + p(t) \cdot y' + q(t) \cdot y = 0 \qquad y(t^*) = m, \quad y'(t^*) = n \quad \text{for} \quad t \in I \quad \text{and} \quad t^* \in I$$

By Theorem 1.5, we have

$$\forall t_0 \in I \Big[\Big(W(y_1, y_2)(t_0) \neq 0 \Big) \leftrightarrow \forall m, n \Big(\exists c_1, c_2 \in \mathbb{R} \text{ s.t. } \mathcal{S}_{\texttt{IVP}}[t_0, m, n] = c_1 y_1 + c_2 y_2 \Big) \Big]$$

Instantiate m, n with $y(t_0), y'(t_0)$ for all $y \in S$:

$$\forall t_0 \in I\Big[\Big(W(y_1,y_2)(t_0) \neq 0\Big) \;\leftrightarrow\; \forall y \in S\Big(\exists c_1, c_2 \in \mathbb{R} \text{ s.t. } \mathcal{S}_{\texttt{IVP}}[t_0, y(t_0), y'(t_0)] = c_1 y_1 + c_2 y_2\Big)\Big]$$

Since y satisfies the ODE and initial conditions itself, $\mathcal{S}_{\texttt{IVP}}[t_0, y(t_0), y'(t_0)]$ is identical to y. Thus we obtain

$$\forall t_0 \in I\Big[\Big(W(y_1,y_2)(t_0) \neq 0\Big) \;\leftrightarrow\; \forall y \in S\Big(\exists c_1, c_2 \in \mathbb{R} \text{ s.t. } y = c_1 y_1 + c_2 y_2\Big)\Big] \tag{5}$$

Now we can apply the following corollary from first-order formal logic:

$$\Big(\exists x.R(x)\Big) \;\wedge\; \forall x.\Big(R(x) \to (P(x) \leftrightarrow Q)\Big) \quad\implies\quad \Big(\exists x.(R(x) \wedge P(x))\Big) \leftrightarrow Q$$

(Note that Q does not depend on x). Identifying $R(x)$ with $x \in I$, $P(x)$ with $W(y_1,y_2)(x) \neq 0$, and noting I is non-empty so that $\exists t_0 \in I$, we can transform eq. (5) into

$$\Big(\exists t_0 \in I,\; W(y_1,y_2)(t_0) \neq 0\Big) \;\leftrightarrow\; \forall y \in S\Big(\exists c_1, c_2 \in \mathbb{R} \text{ s.t. } y = c_1 y_1 + c_2 y_2\Big)$$

This completes the proof. $\square$

Theorem 1.7 (Existence of Fundamental Set of Solutions)**.**

If $p(t)$ and $q(t)$ are continuous on an open interval I, then there must exist fundamental sets of solutions for the following ODE:

$$y'' + p(t) \cdot y' + q(t) \cdot y = 0 \quad \text{for} \quad t \in I \tag{6}$$

A FSS can be constructed as follows: Pick any $t_0 \in I$, and let y_1 be the solution of eq. (6) with initial conditions

$$y_1(t_0) = 1, \quad y_1'(t_0) = 0$$

Let y_2 be the solution of eq. (6) with initial conditions

$$y_2(t_0) = 0, \quad y_2'(t_0) = 1$$

Then, y_1 and y_2 form a FSS of eq. (6).

Proof.

Theorem 1.1 (Existence and Uniqueness Theorem) guarantees the existence of y_1 and y_2. By the definition of Wronskian, we have

$$W(y_1,y_2)(t_0) = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \neq 0$$

By the corollary of Abel's Theorem, $W(y_1,y_2)(t) \neq 0$ for all $t \in I$. y_1 and y_2 form a FSS of eq. (6). $\square$

2 Homogeneous Equations with Constant Coefficients